Lecture 1: Elements of Probability

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Topics of this lecture:

1. Random Experiment
2. Sample Space
3. Event
4. Probability

1) Random Experiment:

An experiment which exact outcome is unknown
But set of all possible outcomes is known
Is known as random experiment.

Example:

1) Tossing a coin
2) Throwing a die
3) Drawing a card
4) Guess a number

$S = \{H, T\} \rightarrow |S| = 2$

$S = \{1, 2, 3, 4, 5, 6\} \rightarrow |S| = 6$

$S = \{1, 2, 3 \ldots 51, 52\} \rightarrow |S| = 52$

$S = \{1, 2, \ldots 9, 10\} \rightarrow |S| = 10$

2) Sample Space(S):

Set of all possible outcomes of a random experiment
is known as sample space.
It is denoted by "S".

Example:

1.Tossing a coin **S = {Head, Tail}** |S| = 2	**2.Throwing a die** **S = {1,2,3,4,5,6}** |S| = 6
3.Drawing a card **S = {1, 2, 3,……51, 52}** |S| = 52	**4.Guess a number between 1 to 10** **S = {1, 2, 3,……9, 10}** |S| = 10

3) Event(E):

Any subset of sample space is known as event.
It is denoted by "E".

$P = \frac{|E|}{|S|}$ → no. of favourable outcomes

$|S|$ → Total no. of outcomes.

Example 1:

- **Random Experiment : Throwing a die**
- **Sample Space (S) : {1,2,3,4,5,6}** $|S| = 6$
- **Event (E) : Probability of getting a 6.**

$P(E) = \frac{1}{6} = 16.66\%$

$$\boxed{P(E) = \frac{|E|}{|S|}}$$

Example 2:

$P(E) = \frac{|E|}{|S|} = \frac{2}{6} = \frac{1}{3} = 33.33\%$

Q. There are 6 balls in a bucket
3 balls are red
2 balls are green
1 ball is blue
You have to pick 1 green ball

Here-

- **Sample Space (S) : {O O O O O O}** $|S| = 6$
- **Event (E) : Getting 1 green (O) ball.**

4) Probability P(E):

Probability is always measured of an event.

Probability is a measure of likelihood of an event to occur

Probability of event E:

$$P(E) = \frac{\text{No. of favourable outcomes}}{\text{Total no. of outcomes in Sample Space}}$$

$$\boxed{P(E) = \frac{|E|}{|S|}}$$

0

Probability of getting one green ball is:

$$\mathbf{P(E) = \frac{2}{6} = 33.33\%}$$